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Faculty of Computer Science and Engineering

CCE 310 : Software Development Project-I

Project title : KitchenCraft

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KitchenCraft: A Cross-Platform Culinary Management Application

Introduction

In an increasingly digital world, the convenience of managing daily tasks through applications has become paramount. Cooking and meal preparation, fundamental aspects of daily life, often benefit significantly from organized digital tools. KitchenCraft is envisioned as a cutting-edge culinary management application built on the Flutter framework, offering a seamless and integrated experience across mobile, web, and desktop platforms. The application is designed to be a user's ultimate kitchen companion, simplifying everything from grocery shopping to daily meal planning. By leveraging modern technologies like WebSockets for real-time synchronization, SQLite for structured data storage, Hive for efficient offline caching, and potentially advanced AI services for intelligent assistance, KitchenCraft aims to provide a robust, reliable, and intuitive platform. Its Material Design-inspired UI ensures a consistent and polished look and feel across all supported devices, making it accessible and enjoyable for a wide range of users. This report details the proposed architecture, methodology, and features of the KitchenCraft application.

Objective

The objective of the KitchenCraft project is to develop a self-hosted, cross-platform application for managing grocery lists, recipes, meal plans, and expenses, with real-time collaboration and an AI assistant to enhance user experience.

The app aims to:

- Provide a seamless, native experience on mobile, web, and desktop platforms.
- Enable real-time syncing of grocery lists across multiple users. Support offline functionality for viewing and modifying data.
- Facilitate recipe creation, sharing, and integration with grocery lists and meal plans.
- Track household expenses and provide budgeting insights.
- Integrate an AI assistant for personalized recipe suggestions and grocery optimization.

Problem Statement

Modern individuals often face challenges in managing their culinary activities efficiently. These challenges include:

Disorganized Grocery Shopping: Forgetting items, inefficient list management, and lack of real-time synchronization among household members.

Recipe Overload and Accessibility: Difficulty in organizing personal recipes, easily adding ingredients to shopping lists, and accessing recipes offline.

Lack of Structured Meal Planning: Haphazard meal planning leading to last-minute decisions, food waste, and repetitive meals.

Untracked Culinary Expenses: Difficulty in monitoring spending related to groceries and food, hindering budgeting efforts.

Information Overload: Searching for cooking inspiration or specific culinary information across various platforms can be time-consuming and fragmented.

Platform Fragmentation: Relying on multiple, disparate applications for different culinary needs across various devices.

KitchenCraft seeks to address these problems by providing a unified, intuitive, and intelligently assisted platform.

Related Work

Several existing applications address various aspects of culinary management. Understanding their features and limitations provides crucial context for highlighting KitchenCraft's unique value proposition and integrated approach.

1. Grocery List Apps (e.g., AnyList, OurGroceries):

How they work: These applications primarily focus on creating and managing shopping lists. Users can add items, categorize them, and often share lists with family members for collaborative input. Synchronization is a core feature, allowing multiple users to update the same list in real-time.

Limitations: While excellent for list sharing and basic synchronization, they generally lack deep integration with other culinary functionalities. Users cannot typically link list items directly to recipes, schedule them in meal plans, or track

associated costs. Their scope is narrow, requiring users to switch to other apps for broader meal management.

2.Recipe Apps (e.g., Paprika, Mealime):

How they work: These apps serve as digital recipe boxes, allowing users to store, organize, and discover recipes. Features often include importing recipes from websites, creating personal recipes, and providing cooking instructions. Some may offer basic ingredient list generation from recipes, which can then be manually added to a separate grocery list.

Limitations: Their primary focus remains on recipes. While some offer basic meal planning features, they often fall short on real-time grocery list syncing, making collaborative shopping cumbersome.

3.Meal Planning Apps (e.g., Plan to Eat, BigOven):

How they work: Designed to help users schedule meals for specific days or weeks. They provide calendars or planners where users can assign recipes. Often, these apps can generate a consolidated grocery list based on the ingredients of the planned meals. Some may offer a database of recipes or allow users to add their own.

Limitations: While strong in planning, their integration with real-time grocery list updates is often limited, and they might not support true real-time syncing across devices. Their recipe management capabilities might be less robust than dedicated recipe apps, and features like detailed expense tracking or AI-driven meal suggestions are generally not core offerings.

4.AI-Powered Cooking Apps (Emerging):

How they work: This is a newer category where applications leverage artificial intelligence or machine learning to provide recipe recommendations, answer cooking-related questions, suggest ingredient substitutions, or even generate new recipes based on available ingredients.

Limitations: While innovative in their use of AI, these platforms are often narrowly focused on the AI interaction itself. They typically do not offer a full suite of features like robust real-time grocery list synchronization, comprehensive recipe management with offline access, detailed meal planning, or integrated

expense tracking. They often serve as a specialized tool rather than an all-in-one solution.

KitchenCraft differentiates itself by offering a truly holistic solution that integrates real-time synced grocery lists, robust recipe and meal plan management, precise expense tracking, and an intelligent, interactive AI assistant, all within a single, consistent, and cross-platform Flutter application. This comprehensive approach aims to reduce the need for multiple disparate applications, minimize context switching, and provide a more cohesive and empowering user experience for all culinary activities.

Proposed Features

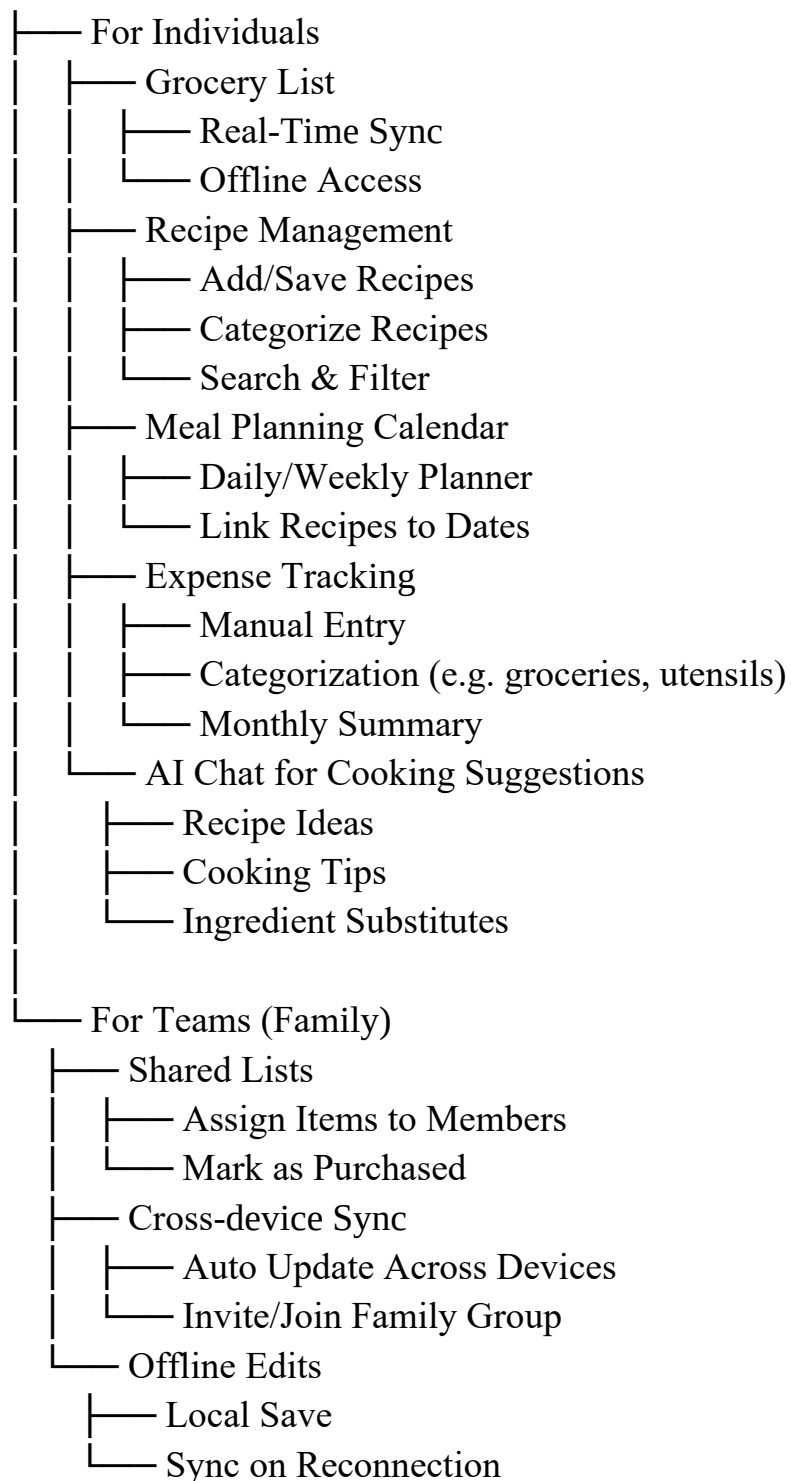
For Individuals:

- Grocery List (sync + offline)
- Recipe Management
- Meal Planning Calendar
- Expense Tracking
- AI Chat for Cooking Suggestions

For Teams (Family):

- Shared Lists
- Cross-device Sync
- Offline Edits

Tree structure :
KitchenCraft App



Brief Technology Description

The KitchenCraft application will leverage a modern and robust technology stack to ensure performance, cross-platform compatibility, and a rich user experience.

Brief Technology Description

KitchenCraft leverages modern technologies for robust functionality, with Firebase serving as the primary database for scalable, real-time data management:

Frontend: Flutter (Dart) for cross-platform development, providing native performance with a single codebase for mobile (iOS/Android), web, and desktop.

Backend/Database: Firebase Firestore as the core NoSQL database for storing recipes, groceries, meal plans, expenses, and family groups, enabling real-time synchronization and offline persistence. Firebase Auth handles user authentication, while Firebase Storage manages any uploaded images (e.g., recipe photos).

Offline Support: Hive for lightweight local caching of frequently accessed data (e.g., grocery lists, chat history), ensuring partial offline functionality. SQLite (via sqflite plugin) supplements for relational queries, such as detailed expense logs or recipe ingredients.

Real-Time Communication: WebSockets (via `socket_io_client`) for instant updates in shared lists and family collaboration, broadcasting changes across devices.

Backend Server (AI Proxy & WebSockets): A Node.js/Express server acts as an intermediary for secure WebSocket connections and proxies AI requests to prevent key exposure in the client.

AI Integration: Google Gemini API (primary) for natural language processing and recipe suggestions, with fallback to local LLM (e.g., via Ollama) for offline use. The backend proxy routes queries with user context (e.g., grocery list).

State Management: Provider for efficient UI updates and global state (e.g., user data, grocery list).

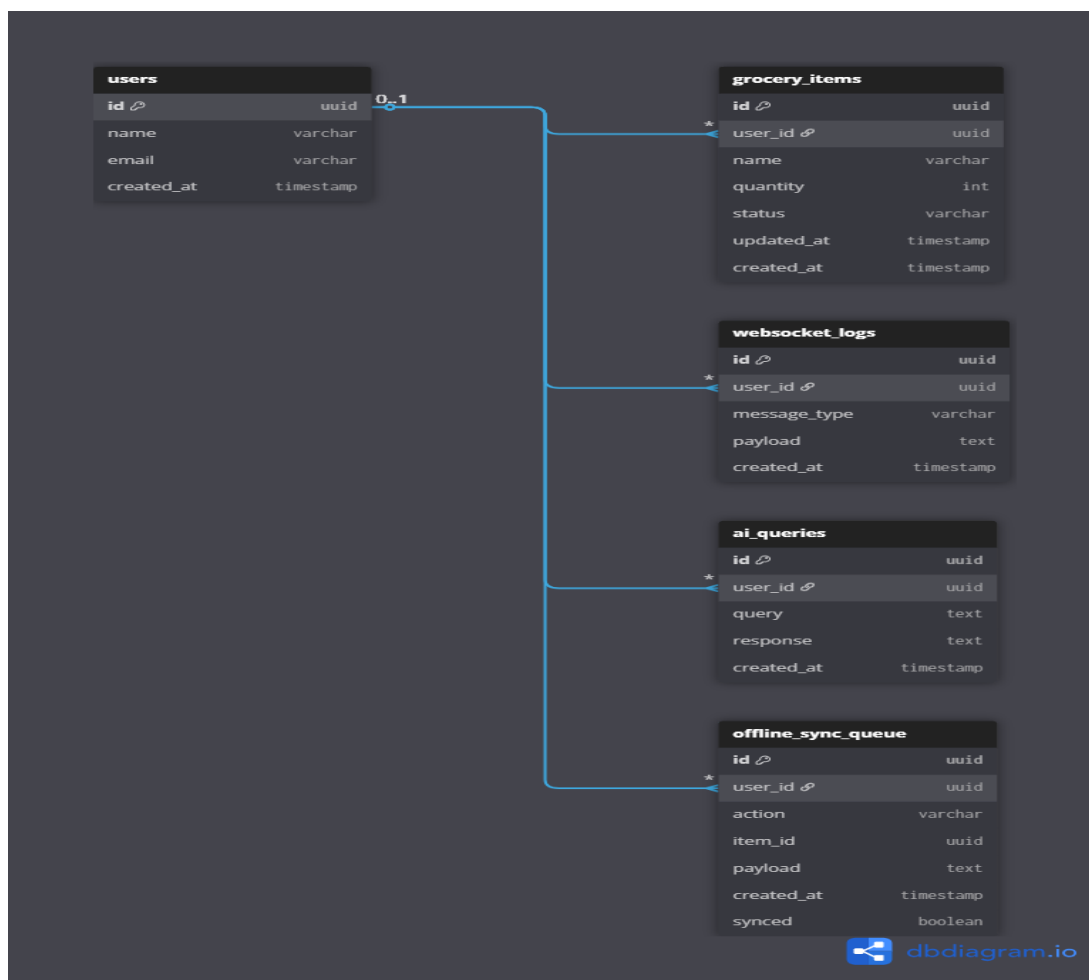
UI Framework: Material Design for intuitive, responsive interfaces across platforms.

Working Process

The development of KitchenCraft will follow an iterative and agile methodology, allowing for flexibility, continuous integration, and rapid feedback loops. The process will involve distinct phases including planning, design, development, testing, and deployment.

1. Proposed ER Diagram: The Entity-Relationship (ER) Diagram below illustrates the proposed structure of the core data entities within the KitchenCraft application and their relationships. This will primarily inform the SQLite database schema.

Figure 2: KitchenCraft Proposed ER Diagram



Explanation of Entities and Relationships:

USER: Represents an application user. Central to all personalized data.

RECIPE: Stores user-created recipes.

Relationship with USER: One User can create many Recipes.

INGREDIENT: A master list of ingredients.

RECIPE_INGREDIENT: An associative entity that links Recipes to Ingredients, capturing the specific quantity and unit for each ingredient within a recipe (many-to-many relationship).

GROCERY_LIST_ITEM: Individual items on a user's grocery list.

Relationship with USER: One User has many Grocery List Items.

MEAL_PLAN_ENTRY: Records a recipe scheduled for a specific date.

Relationship with RECIPE: One Recipe can be part of many Meal Plan Entries.

Relationship with USER: One User can have many Meal Plan Entries.

EXPENSE: Records a single expense transaction.

Relationship with USER: One User logs many Expenses.

2. Workflow of API Integration

The API workflow diagram primarily illustrates the real-time synchronization of the Grocery List and the interaction with the AI Assistant, both relying on a backend server.

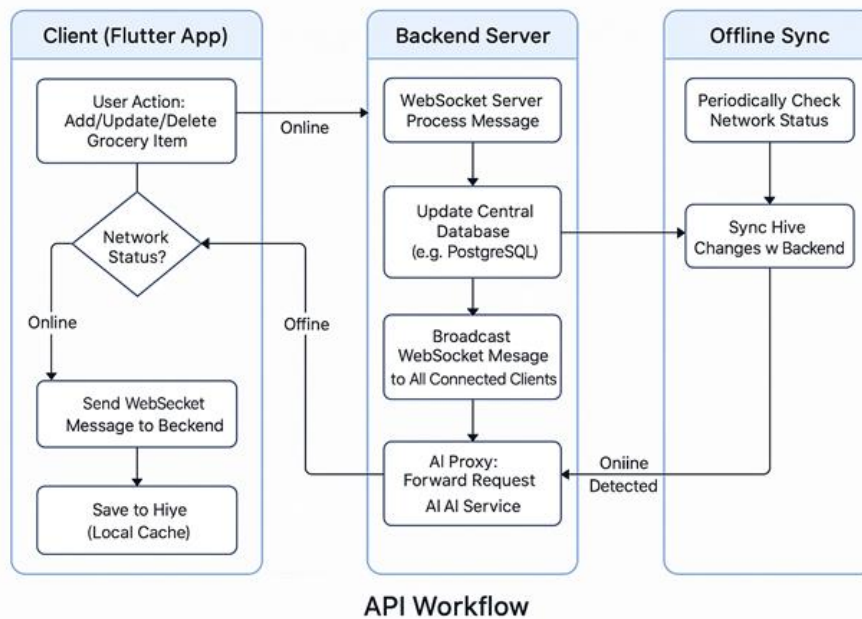


Figure 3: API Workflow Diagram

Explanation of Workflow:

Grocery List Sync:

When a user modifies a grocery item in the Flutter app, it first checks network status.

Online: A WebSocket message is immediately sent to the Backend Server. The server updates its central database, then broadcasts the change to all connected clients (including the originating client), ensuring real-time consistency.

Offline: Changes are saved locally to Hive. When network connectivity is restored, a background process syncs these queued changes with the backend, which then broadcasts to other clients.

Incoming WebSocket messages from the backend update the local Hive cache and the UI.

AI Assistant Interaction:

A user query from the Flutter app is sent via an HTTP request to the Backend Server's AI Proxy.

The AI Proxy securely forwards this request to the actual External AI Service (e.g., xAI API, Google Gemini, or a locally hosted LLM).

The AI Service processes the query and sends a response back to the AI Proxy.

The AI Proxy then relays this response back to the Flutter app, which displays it in the chat interface.

3. Flow Chart (General App Logic)

This flowchart provides a high-level overview of the main user flows and decision points within the KitchenCraft application.

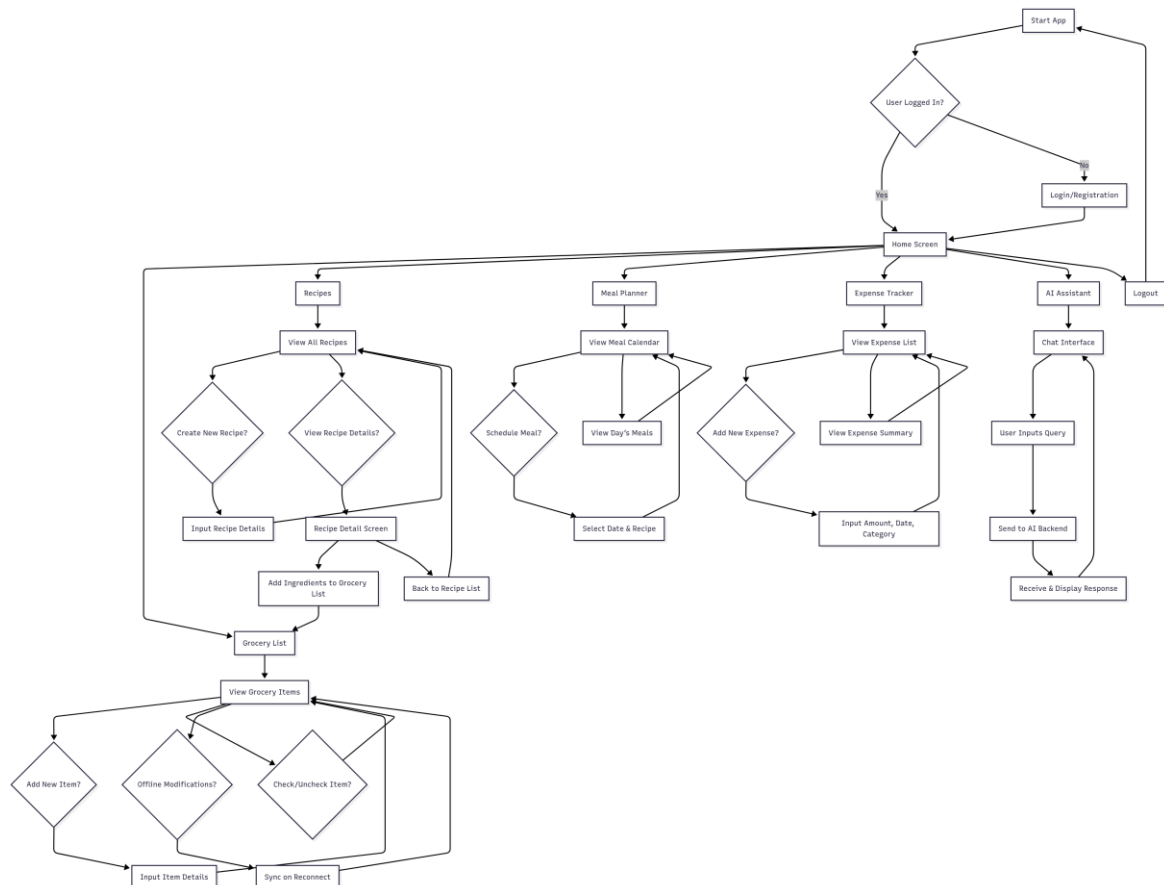


Figure 4: General Application Flow Chart

UI Mockup

KitchenKraft

Login to your account

[Forgot Password?](#)

Login

or



Don't have an account?
[Sign up](#)

KitchenKraft

Sign Up



email@example.com

at least 8 characters

Sign Up

Already have an account?
[Log In](#)

KitchenKraft

Good morning, [User Name]!
Upcoming tasks are liave bufertion,
Quick stats for Curee right.

Grocery List

☐ Milk

Swipe for options >

☐ Eggs

3x >

☐ Bread

3x >

☐ Piwe

☐ Bread

1 lb >

☐ Apples

1 lb >

Quick Access Recipes

Meal Plans

Sequs

Grocery

Recipes

Plans

Expenses

Gantt Chart (Placeholder)

Task Description	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9	Week 10-11	Week 12
Planning & Design	✓						
Flutter Setup & Grocery List UI		✓					
Recipe & Meal Planning UI		✓	✓				
Expense Tracking & Offline Sync			✓				
WebSocket Sync & Authentication				✓			
Recipe Sharing & Collaboration				✓			
AI Assistant Integration				✓			
Security & Analytics					✓		
Testing & Optimization					✓		
Deployment (Mobile/Web/Desktop)							
✓							

Results and Limitations

This section will be fully elaborated upon project completion.

1. Expected Results

Upon successful completion, KitchenCraft is expected to deliver:

- A fully functional, responsive, and visually appealing application across Android, iOS, Web, and Desktop platforms.
- Seamless real-time grocery list synchronization across multiple devices.
- Robust offline capabilities for grocery lists, recipes, meal plans, and expenses.
- Intuitive recipe creation, management, and one-tap ingredient addition to the grocery list.
- An effective meal planning calendar integrated with user recipes.
- A straightforward expense tracking system.
- An intelligent AI assistant capable of responding to culinary queries and providing suggestions.
- A scalable architecture that allows for future feature additions and improvements.

2. Current Limitations (Anticipated)

During development and initial deployment, the following limitations are anticipated:

- **AI Service Cost/Rate Limits:** Reliance on external AI APIs may incur costs or be subject to rate limits, potentially impacting user experience or requiring a paid subscription model.
- **Offline Conflict Resolution Complexity:** While basic offline support is planned, advanced conflict resolution (e.g., when two users simultaneously modify the same grocery item offline) might be rudimentary in the initial version.
- **Initial Feature Scope:** Features like advanced charts for expense tracking, collaborative recipe editing (beyond grocery list syncing), or highly personalized AI suggestions might be beyond the scope of the initial release and considered for future iterations.
- **Backend Scalability:** The initial backend infrastructure might be designed for a moderate number of users. Scaling for a very large user base would require further optimization and infrastructure investment.
- **Platform-Specific Nuances:** Despite Flutter's cross-platform nature, minor platform-specific behaviors or UI adjustments might be necessary, requiring additional development effort for each platform.

Conclusion

KitchenCraft represents an ambitious yet achievable project to create a unified and intelligent culinary management application. By strategically leveraging

Flutter's cross-platform capabilities, robust local storage solutions like Hive and SQLite, and modern communication protocols like WebSockets, the application aims to solve common pain points in home cooking and meal preparation. The integration of an AI assistant further elevates the user experience by offering intelligent guidance and inspiration.

While challenges related to real-time synchronization, offline data handling, and AI integration complexity exist, the proposed architecture and methodology provide a clear roadmap for addressing these. Upon completion, KitchenCraft is poised to become an indispensable tool for anyone looking to streamline their kitchen activities, offering unparalleled convenience and smart assistance in one polished package. This project will serve as a strong demonstration of modern cross-platform application development principles and intelligent system integration.

References (Placeholder)

- [Flutter Official Documentation](#)
- [suspicious link removed]
- [sqflite Plugin Documentation](#)
- [WebSocket Protocol Specifications \(if applicable for backend development\)](#)
- [Chosen AI Service API Documentation (e.g., xAI, Google Gemini, OpenAI)]
- [Any specific articles or academic papers on real-time data synchronization or offline-first app design.]
- Android studio: <https://developer.android.com/studio>.

Progress Table

No	Date	Objective (Work Done)
1		Setup Flutter environment & created project kitchen_craft
2		Designed Login & Signup UI Pages

3		Connected project with Firestore (google-services.json, Gradle setup, dependencies)
4		Implemented Firestore Login & Signup functionality
5		Added Grocery List (sync + offline) module
6		Developed Recipe Management (Add/Save/Search recipes)
7		Integrated Meal Planning Calendar (Daily/Weekly planner)
8		Implemented Expense Tracking (manual entry + monthly summary)
9		Added AI Chat for Cooking Suggestions (recipe ideas, tips, substitutes)
10		Completed Family/Team Features (Shared Lists, Cross-device Sync, Offline Edits) & delivered full KitchenCraft App